

Lab 09 Report

Mini Project 01: Two-Stage Miller OTA

1. Part 1: g_m/I_D Design Charts

1.1 Design Charts (NMOS & PMOS, $V_{DS} = V_{DD}/3$, $L = 0.18\mu\text{m}$: $0.5\mu\text{m} : 2\mu\text{m}$ $I_D = 10\mu\text{A}$)

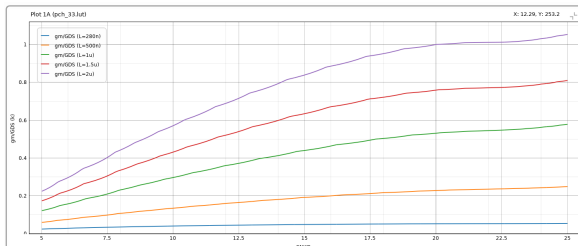


Figure 1: PMOS g_m/g_{ds} vs. g_m/I_D

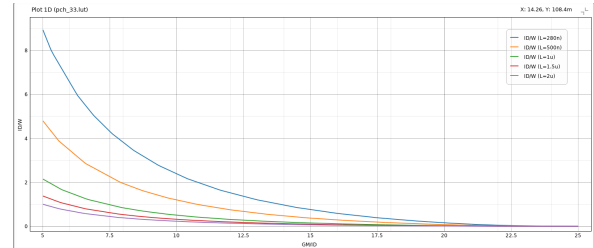


Figure 2: PMOS I_D/W vs. g_m/I_D

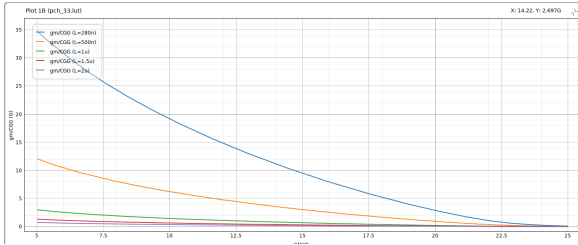


Figure 3: PMOS g_m/c_{gg} vs. g_m/I_D

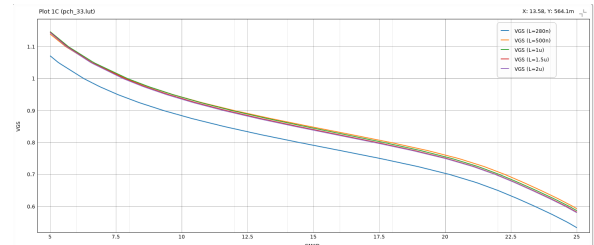


Figure 4: PMOS V_{GS} vs. g_m/I_D

Figure 5: PMOS lookup table characteristic curves across different channel lengths.

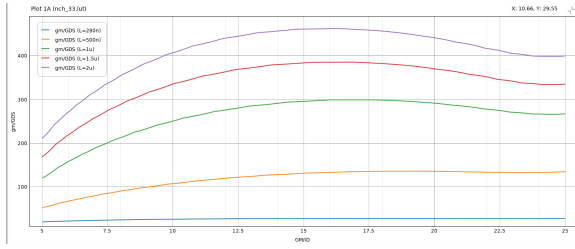
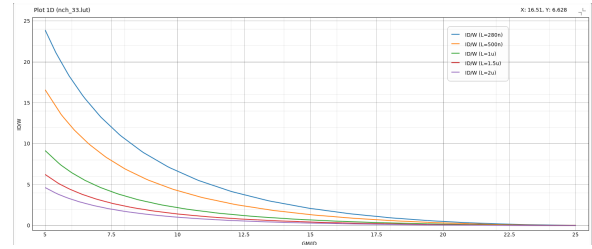
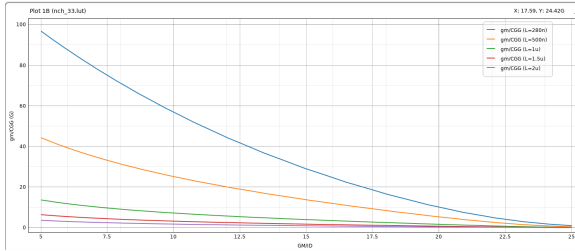
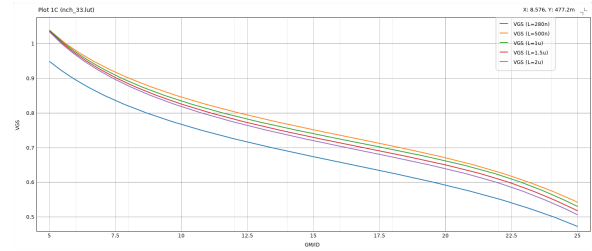
Figure 6: NMOS g_m/g_{ds} vs. g_m/I_D Figure 7: NMOS I_D/W vs. g_m/I_D Figure 8: NMOS g_m/c_{gg} vs. g_m/I_D Figure 9: NMOS V_{GS} vs. g_m/I_D

Figure 10: NMOS lookup table characteristic curves across different channel lengths.

2. Part 2: Two-Stage Miller OTA Design

2.1 Specifications

Parameter	Value
Technology	ADT45nm
Supply voltage V_{DD}	1.8
Static gain error	$\leq 0.05\%$
CMRR @ DC	≥ 74 dB
Phase margin	$\geq 70^\circ$
Total current consumption	$\leq 60 \mu A$
CMIR – high	$\geq 0.8V$
CMIR – low	≤ 0.2 V
Output swing	0.2-1.6V
Load C_L	5 pF
Buffer rise time (10%–90%)	≤ 70 ns
Slew rate	5 V/ μs

Table 1: Target OTA specifications

2.2 Input Pair Type Justification

Why NMOS or PMOS input pair for each stage?

I would use PMOS input stage since $CMIR_{LOW}$ is closer to ground to keep the PMOS in saturation. I would have used a rail-to-rail input stage but it is not worth it as $CMIR_{HIGH}$ is far from VDD.

2.3 Hand Analysis: Design Procedure

Step 1–2: Test bench & C_C

Due to current and area constraints. I chose to make $V_{SB} = 0$ for all devices.

Substrate Connections and the Body Effect

- **Why it is impossible for all NMOS:** In a standard CMOS process, all NMOS devices share a single, global p-substrate. To prevent forward-biasing the internal PN junctions, this entire substrate must be tied to Ground. If an NMOS has a floating source (e.g., a differential pair tail), tying its bulk to its source would short the global substrate to that node. This forces $V_{SB} > 0$, resulting in an unavoidable body effect.
- **Why it is possible for PMOS:** PMOS devices are fabricated inside individual, isolated n-wells. This physical isolation allows the bulk (n-well) of each PMOS to be tied directly to its respective source, ensuring $V_{SB} = 0$ without affecting other devices.
- **Layout Solution for NMOS:** To achieve $V_{SB} = 0$ for an NMOS with a floating source, the layout must utilize a **twin-well or triple-well (deep n-well)** technology. This isolates the NMOS inside a “private” p-well, separating its local bulk from the global substrate so it can be safely tied directly to its source.

$$I_{B1} = 10 \mu\text{A} \quad (1)$$

$$I_{B2} = 40 \mu\text{A} \quad (2)$$

$$I_{BiasRz} = 5 \mu\text{A} \quad (3)$$

Step 3: Slew Rate $\rightarrow$ First Stage Current

$$SR = \frac{I_{B1}}{C_C} = 5V/\mu s \quad (4)$$

$$I_{B1} = SR \cdot C_C \approx 10\mu A \quad (5)$$

$$I_{B2} = I_{total} - I_{B1} = 40\mu A \quad (6)$$

Step 4: Unity Gain Frequency from Rise Time

To get $I_{B1} = 10\mu A$. I chose $C_c = 0.4C_L = 2pF$. To account for parasitics. I chose $C_c = 1.75pF$

$$t_{rise} = 2.2 \tau \quad (7)$$

$$\omega_u = \frac{0.35}{t_{rise}} = 5MHz \quad (8)$$

$$g_{m1,2min} = \omega_u C_C = 54.97\mu S \quad (9)$$

Step 5: Closed-Loop Gain Error $\rightarrow$ Required Open-Loop Gain

$$A_{v,CL} \approx 1 - \frac{1}{A_{v,OL}} \quad (10)$$

$$\epsilon_s = \left| \frac{A_{v,CL} - 1}{1} \right| \times 100 \quad (11)$$

$$A_{v,OL} \geq \frac{1}{\epsilon_s/100} \quad (12)$$

From that $A_{v,OL} \geq 2000$

Step 6: Phase Margin

To meet the PM spec

$$\frac{3.5g_{m1,2}}{C_c} = \frac{g_{m_{secondstage}}}{C_L}$$

Minimum g_m of the second stage input is $549.7\mu S$

Step 7: Gain Split Between Stages

Q: Why is the first stage assigned larger gain (twice the second stage, i.e. 6 dB difference)?

Because I want output swing from the second stage and gain is inversely proportional to V^* .

Step 8: Design of input pair

Assume $V_{\text{tail}}^* = 150\text{ mV}$ (sweet spot) and from the $\text{CMIR}_{\text{high}}$ spec. We get $V_{SG} < V_{DD} - V_{\text{tail}}^* - 0.8 \approx 0.85\text{ V}$. So we need moderate inversion.

Furthermore, We need gain from first stage of 64 so I will set $\frac{g_m}{g_{ds}} > 128$.

Assuming $\frac{g_m}{I_D} = 16$ so, $g_{m,\text{input}} = 80\text{ }\mu\text{S}$.

▼ LUT Settings

LUT ?

Corner ☐ All ?

Temp (°C) ☐ All ?

Frequency ?

ID ?

gm/ID ?

gm/gds ?

VDS ?

VSB ?

Stack ?

Results:

	Name	TT-27.0
3	L	410n
4	W	13.32u
5	VDB	600m
6	VGB	824.5m
7	VGS	824.5m

Figure 11: Sizing for Input stage

Step 9: Design of Active Load

$$g_{ds} = \frac{I_D}{V_A} < 629.72 \text{ nS} \rightarrow r_o \geq 1.588 \text{ M}\Omega$$

$$V_A > 7.94$$

Also, From $\text{CMIR}_{\text{low}} = V_{GS,\text{load}} - v_{thp}$ spec $V_{GS,\text{load}} \leq v_{thp} + 0.2 = 978.1 \text{ mV}$.

LUT Settings

LUT: nch_33

Corner: TT

Temp (°C): 27.0

Frequency: 1

ID: 5u

gm/ID: 16

VA: 8

VDS: 0.6

VSB: 0

Stack: 1

Results:

Name	TT-27.0
3 L	490n
4 W	4.6u
5 VDB	600m
6 VGB	737.1m
7 VGS	737.1m

Figure 12: Sizing of active load

Sizing the tail current source

$$A_{vCM} = \frac{1}{2g_{m,\text{input}}R_{SS}} < 0.0124$$

$$r_o > 513.75 \text{ k}\Omega$$

Assume $\frac{g_m}{I_D} = 12$

A good practice for current mirror sizing is to size for $\lambda < 0.1$ and $\sigma(I_{out})/I_{out} < 2\%$

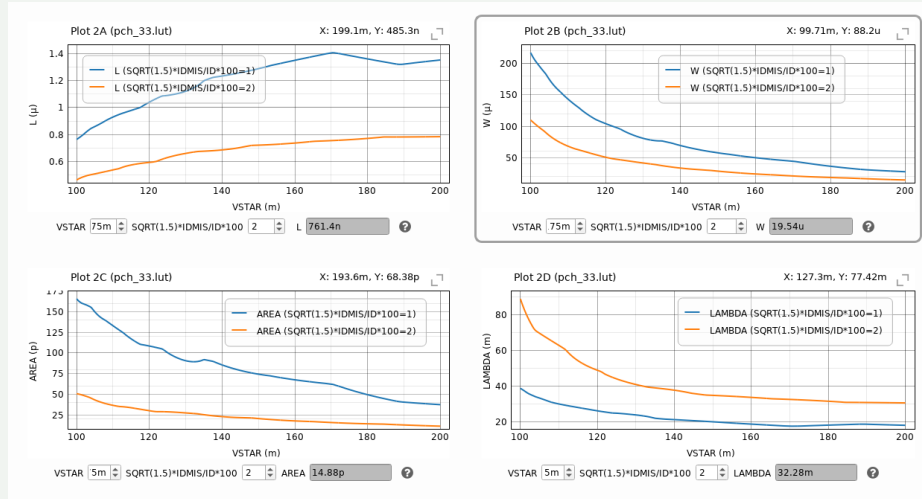


Figure 13: Sizing of tail current source at $I_D = 10\mu A$ and $V_{DS} = 0.9$

▼ LUT Settings

LUT:

Corner: ☐ All

Temp (°C): ☐ All

Frequency:

ID:

Vstar:

L:

VDS:

VSB:

Stack:

Results:

	Name	TT-27.0
4	W	19.52u
5	VDB	900m
6	VGB	914.6m
7	VGS	914.6m
8	VDS	900m

Figure 14: ADT SA design for tail current source

[22-08-2026 18:54:44] - Info: [Sizing Assistant] The resultant point 'LAMBDA = 32.36m'
 [22-08-2026 18:55:01] - Info: [Sizing Assistant] The resultant point 'sqrt(1.5)*idmis/id*100 = 2.003'

Figure 15: Lambda and Mismatch estimates from sizing

Step 11: Second stage input

$$r_{o,\text{load}} = 773.3\text{k}\Omega$$

Assume $\frac{g_m}{I_D} = 16$

$$g_m = 640 \mu\text{S}$$

$$r_o > 53.45 \text{ k}\Omega$$

(from spec $A_{v2} > 32$) however this will result in excessive area.

Try $\frac{g_m}{I_D} = 10$

Results:	
Name	TT-27.0
3 L	370n
4 W	7.28u
5 VDB	900m
6 VGB	813m
7 VGS	813m

Figure 16: Sizing of input second stage

However, because $V_{GS,second} \neq V_{GS,load}$, a systematic DC offset was introduced at the gate of the second stage. This error voltage was amplified by the high stage gain, driving the output to the V_{DD} rail and forcing the second-stage load transistor out of saturation.

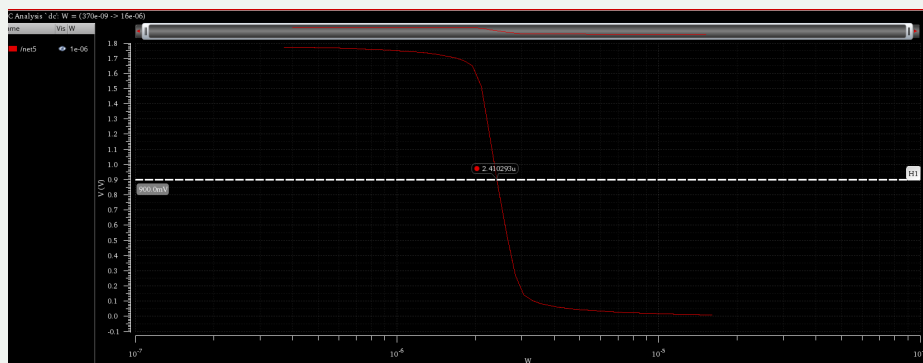
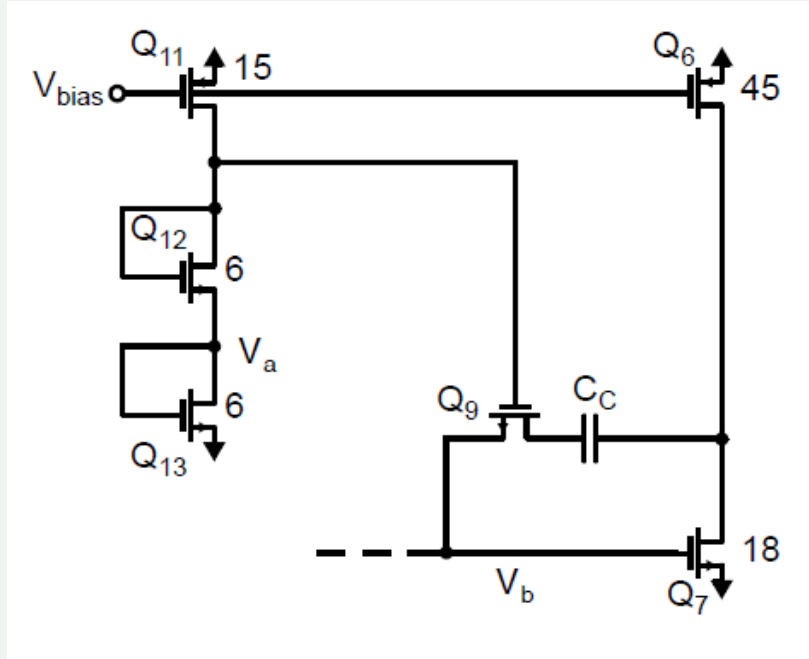


Figure 17: Sweeping W of NMOS input (with m=8) to achieve $V_{out}=V_{DD}/2$

Step 12: R_z SelectionFigure 18: Bias Circuit for R_z **1. Frequency Compensation ($C_C = 1.75$ pF):**

From the small-signal model, the second non-dominant pole and the right-half-plane (RHP) zero are approximated as:

$$\omega_{p2} = \frac{g_{m7}}{C_1 + C_2} \approx \frac{g_{m7}}{C_L} \quad (13)$$

$$\omega_z = \frac{-1}{C_C \left(\frac{1}{g_{m7}} - R_z \right)} \quad (14)$$

To make the RHP zero to infinity (as it will increase GX and decrease PX, worsening PM), an NMOS transistor operating in the triode region (M_9) is used as the nulling resistor (R_z) to track g_{m7} across variations:

$$R_z = r_{ds9} = \frac{1}{\mu_n C_{ox} \left(\frac{W}{L} \right)_9 V_{ov9}} \quad (15)$$

$$g_{m7} = \mu_n C_{ox} \left(\frac{W}{L} \right)_7 V_{ov7} \quad (16)$$

2. Second Stage & Bias Currents:

The second stage branch (M_7, M_8) is designed to conduct $40 \mu\text{A}$, while the bias branch (M_{11}, M_{12}, M_{13}) conducts $5 \mu\text{A}$.

$$I_{D6,7} = 40 \mu\text{A} \quad (17)$$

$$I_{D11,12,13} = 5 \mu\text{A} \implies \frac{I_{D6,7}}{I_{D11,12,13}} = 8 \quad (18)$$

3. Current Mirror Sizing & Systematic Offset Cancellation:

The current is mirrored from the PMOS pair (M_7, M_{11}). Given the dimensions for the PMOS bias current source M_{11} :

$$\left(\frac{W}{L}\right)_{11} = \frac{9.76 \mu\text{m}}{0.76 \mu\text{m}} \approx 12.84 \quad (19)$$

$$\left(\frac{W}{L}\right)_7 = 102.63 \implies \frac{(W/L)_7}{(W/L)_{11}} = 8 \quad (20)$$

To prevent systematic offset and make $V_a = V_b$, the effective overdrive voltages ($V_{\text{eff}} = V_{GS} - V_{TH}$) of the corresponding transistors must be made equal. Starting from the saturation current equation, V_{eff} is isolated:

$$V_{\text{eff}} = \sqrt{\frac{2I_D}{\mu_n C_{ox} \left(\frac{W}{L}\right)}} \quad (21)$$

Forcing $V_{\text{eff}13} = V_{\text{eff}7}$ ensures that $V_{GS13} = V_{GS7}$. Equating the two roots guarantees the drain voltages align perfectly to kill the offset:

$$\sqrt{\frac{2I_{D7}}{\mu_n C_{ox} \left(\frac{W}{L}\right)_7}} = \sqrt{\frac{2I_{D13}}{\mu_n C_{ox} \left(\frac{W}{L}\right)_{13}}} \quad (22)$$

To ensure the node voltages track correctly across the branches and prevent systematic offset, the NMOS aspect ratios must scale by this exact same current ratio:

$$\frac{(W/L)_9}{(W/L)_{13}} = \frac{I_{D7}}{I_{D13}} = 8 \quad (23)$$

Given the dimensions for the second stage NMOS input M_7 ($W_7 = 19.28 \mu\text{m}$ and $L_8 = 370 \text{ nm}$), we enforce identical channel lengths ($L_{13} = L_{12} = 370 \text{ nm}$) to avoid short-channel variations. This means the width scales linearly:

$$W_{13} = \frac{W_7}{8} = \frac{19.28 \mu\text{m}}{8} = 2.41 \mu\text{m} \quad (24)$$

2.4 Sizing Table

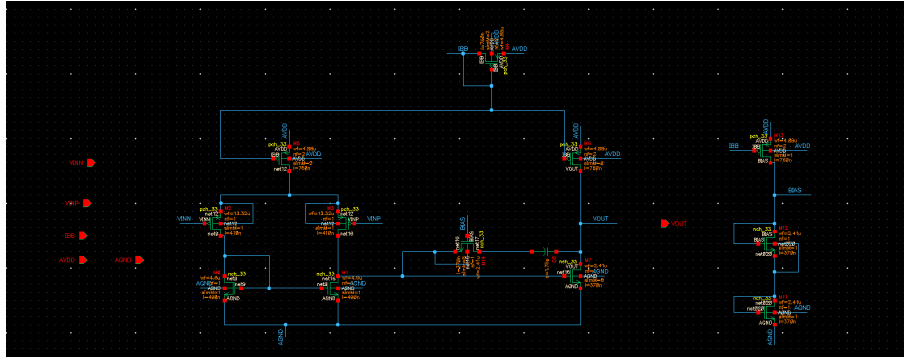


Figure 19: Schematic

Transistor	W (μm)	L (μm)	g_m (μS)	I_D (μA)	g_m/I_D	$ v_{dsat} $ (mV)	$ V_{ov} $ (mV)
M2, M3 (Input Pair)	13.32	0.41	79.59	5	15.92	100.7	46.4
M0, M1 (1st Stage Load)	4.60	0.49	78.25	5	16	101.5	28.2
M5 (Tail Source)	9.76	0.76	113	10	11.3	140.2	127.8
M7 (2nd Stage Input)	19.28	0.37	570.3	39.99	14.3	112.7	50.0
M6 (2nd Stage Load)	78.08	0.76	457.0	39.99	11.4	139.2	126.7
M4 (Bias Diode PMOS)	9.76	0.76	114.3	10.00	11.4	139.2	126.7
M13 (Bias PMOS)	9.76	0.76	55.8	4.85	11.5	138.3	125.7
M11, M12 (Bias NMOS)	2.41	0.37	69.7	4.85	14.4	112.2	49.2
M14 (Nulling Resistor)	2.41	0.37	(Operates in Triode Region)				

Table 2: Transistor sizing and DC operating point summary (W, L, g_m , I_D , g_m/I_D , v_{dsat} , $V_{ov} = |V_{GS}| - |V_{TH}|$)

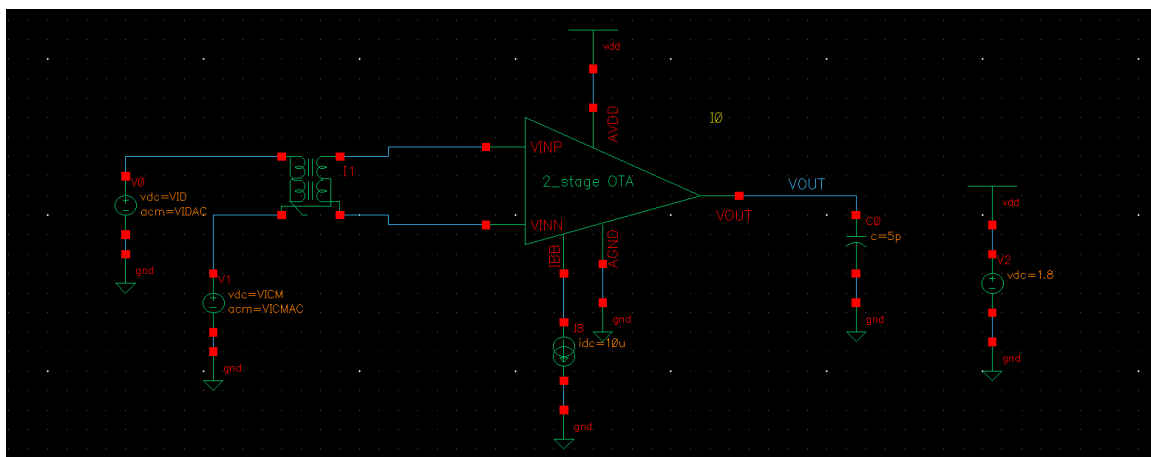


Figure 20: Open Loop Testbench

3. Part 3: Open-Loop OTA Simulation

3.1 DC Operating Point

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1 *****
2 Operating-Point information 'dump_op':
3 *****
4 Operating point computed in info analysis 'dump_op' at T = 27 C.
5
6 Transistor Type Region id vgs vds vdsat vth gm gds
7 -----
8 IO.M0.m0 NMOS sat 4.8859 uA 733.535 mV 733.535 mV 99.2122 mV 709.536 mV 78.0946 uS 558.687 nS
9 IO.M1.m0 NMOS sat 4.8859 uA 733.535 mV 733.535 mV 99.2122 mV 709.536 mV 78.0946 uS 558.687 nS
10 IO.M2.m0 PMOS sat -4.88464 uA -825.27 mV -691.735 mV -98.4954 mV -781.58 mV 78.664 uS 553.371 nS
11 IO.M3.m0 PMOS sat -4.88464 uA -825.27 mV -691.735 mV -98.4954 mV -781.58 mV 78.664 uS 553.371 nS
12 IO.M4.m0 PMOS sat -10 uA -914.072 mV -914.072 mV -139.197 mV -787.329 mV 114.292 uS 314.589 nS
13 IO.M5.m0 PMOS sat -9.76919 uA -914.072 mV -374.73 mV -139.179 mV -787.355 mV 111.601 uS 665.533 nS
14 IO.M6.m0 PMOS sat -39.9851 uA -914.072 mV -902.254 mV -139.196 mV -787.329 mV 457.004 uS 1.26968 uS
15 IO.M7.m0 NMOS sat 39.987 uA 733.535 mV 897.746 mV 112.706 mV 683.517 mV 570.254 uS 8.05801 uS
16 IO.M11.m0 NMOS sat 4.85138 uA 733.817 mV 733.817 mV 112.244 mV 684.571 mV 69.7248 uS 1.03118 uS
17 IO.M12.m0 NMOS sat 4.85138 uA 733.817 mV 733.817 mV 112.244 mV 684.571 mV 69.7248 uS 1.03118 uS
18 IO.M13.m0 PMOS sat -4.85138 uA -914.072 mV -332.367 mV -138.312 mV -788.339 mV 55.8276 uS 390.179 nS
19 IO.M14.m0 NMOS triode 0 A 734.098 mV 0 V 109.622 mV 689.291 mV 0 S 522.298 uS

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Figure 21: DC Operating points at $V_{ICM}=V_{DD}/3$

DC Operating Point Questions

Q: Is the current (and g_m) in the input pair exactly equal?

Yes they are the same.

Q: What is the DC voltage at the output of the first stage? Why?

$V_{GS1} = V_{DS1} = 733.535mV$. The two nodes are virtually connected.

Q: What is the DC voltage at the output of the second stage? Why?

$V_{DS7} = 897.746mV$ because we forced that when sweeping W_7

3.2 Differential Small-Signal AC Analysis

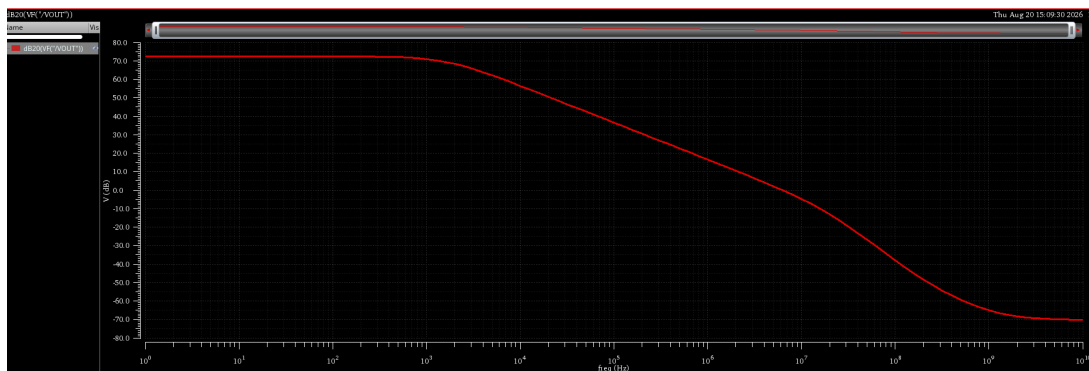


Figure 22: Differential gain (dB) vs. frequency ($V_{IDAC}=1$, $V_{ICMAC}=0$)

$$\begin{aligned}
A_{v,total} &= A_{v1} \times A_{v2} \\
A_{v,total} &= \left(\frac{g_{m3}}{g_{ds1} + g_{ds3}} \right) \left(\frac{g_{m7}}{g_{ds6} + g_{ds7}} \right) \\
A_{v,total} &= \left(\frac{78.66 \mu\text{S}}{558.69 \text{ nS} + 553.37 \text{ nS}} \right) \left(\frac{570.25 \mu\text{S}}{1.270 \mu\text{S} + 8.058 \mu\text{S}} \right) \\
A_{v,total} &= (70.7 \text{ V/V}) \times (61.1 \text{ V/V}) \\
A_{v,total} &= 4320 \text{ V/V} \approx 72.7 \text{ dB}
\end{aligned}$$

Hand Analysis: Exact Dominant Pole with Parasitics

1. Inter-stage Parasitic Capacitance (C_1):

From the operating point dump, the parasitic capacitance at the output of the first stage consists of the drain capacitances of M_1 and M_3 , and the gate capacitance of M_7 :

$$\begin{aligned}
C_1 &= cdd_{M1} + cdd_{M3} + cgg_{M7} \\
C_1 &= 861.1 \text{ aF} + 2.15 \text{ fF} + 23.81 \text{ fF} = 26.82 \text{ fF}
\end{aligned}$$

2. Exact Dominant Pole Calculation:

Using the complete zero-value time constant method, the dominant pole time constant (τ_{p1}) includes both the precise Miller-multiplied capacitance at the first stage and the time constant of the output stage ($C_L = 5 \text{ pF}$):

$$\begin{aligned}
\tau_{p1} &= R_{out1} [C_1 + C_C(1 + A_{v2})] + R_{out2}(C_C + C_L) \\
\tau_{p1} &= (899.2 \text{ k}\Omega) [26.82 \text{ fF} + (1.75 \text{ pF})(1 + 61.13)] + (107.2 \text{ k}\Omega)(1.75 \text{ pF} + 5 \text{ pF}) \\
\tau_{p1} &= (899.2 \text{ k}\Omega)[108.75 \text{ pF}] + (107.2 \text{ k}\Omega)[6.75 \text{ pF}] \\
\tau_{p1} &= 97.79 \mu\text{s} + 0.72 \mu\text{s} \\
\tau_{p1} &= 98.51 \mu\text{s}
\end{aligned}$$

3. Bandwidth Verification:

$$\omega_{p1} = \frac{1}{\tau_{p1}} = \frac{1}{98.51 \mu\text{s}} \approx 10151 \text{ rad/s}$$

$$\begin{aligned}
BW &= f_{p1} = \frac{\omega_{p1}}{2\pi} \\
BW &\approx 1.615 \text{ kHz}
\end{aligned}$$

4. Gain-Bandwidth Product (GBW):

$$\begin{aligned}
GBW &= A_{v,total} \times BW \\
GBW &= 4320 \times 1.615 \text{ kHz} \\
GBW &\approx 6.98 \text{ MHz}
\end{aligned}$$

This aligns perfectly with the standard theoretical Miller approximation for GBW

$$GBW \approx \frac{g_{m1}}{2\pi C_C} = \frac{78.66 \mu\text{S}}{2\pi(1.75 \text{ pF})} \approx 7.15 \text{ MHz}$$

Diff Gain: Hand Calc vs. Simulation

Parameter	Hand Analysis	Simulation
A_o (dB)	72.7	72.687
BW (kHz)	1.615	1.604
GBW (MHz)	6.98	6.91
f_u (MHz)	-	6.527

3.3 Common-Mode Small-Signal AC Analysis

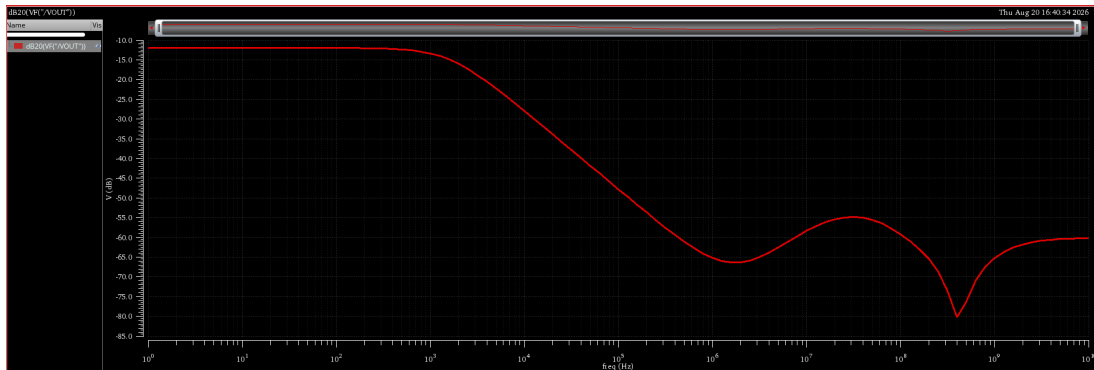


Figure 23: Common-mode gain (dB) vs. frequency (VICMAC=1, VIDAC=0)

Hand Analysis: Common-Mode Gain (A_{CM})

$$A_{CM} \approx \left(\frac{1}{2g_{m1}r_{o5}} \right) \times A_{v2}$$

$$A_{CM} \approx \left(\frac{1}{2g_{m1}r_{o5}} \right) \times g_{m7}(r_{o7} \parallel r_{o6})$$

1. First Stage CM Attenuation:

Using the input pair transconductance ($g_{m1} = g_{m3}$) and the output resistance of the

tail current source (M_5):

$$g_{m1} = 78.66 \mu\text{S}$$

$$r_{o5} = \frac{1}{g_{ds5}} = \frac{1}{665.53 \text{ nS}} \approx 1.502 \text{ M}\Omega$$

$$A_{cm1} \approx \frac{1}{2(78.66 \mu\text{S})(1.502 \text{ M}\Omega)} = \frac{1}{236.3} \approx 0.00423 \text{ V/V}$$

2. Second Stage Gain:

As derived previously, the gain of the second stage is unaffected by common-mode rejection and remains:

$$A_{v2} = g_{m7}(r_{o7} \parallel r_{o6}) = 61.13 \text{ V/V}$$

3. Total Common-Mode Gain:

$$A_{CM} \approx A_{cm1} \times A_{v2}$$

$$A_{CM} \approx (0.00423) \times (61.13)$$

$$A_{CM} \approx 0.258 \text{ V/V} \approx -11.7 \text{ dB}$$

CM Gain: Hand Calc vs. Simulation

Parameter	Hand Analysis	Simulation
A_{cm} (dB)	-11.7	-11.84

(Optional) CMRR

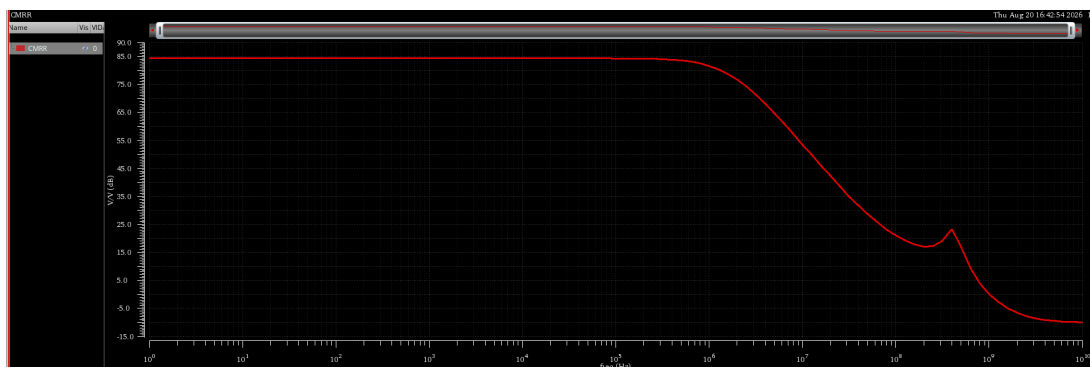


Figure 24: CMRR (dB) vs. frequency, via XF analysis

CMRR: Hand Calc vs. Simulation

Parameter	Hand Analysis	Simulation
CMRR (dB)	84.4	84.527

(Optional) Differential Large-Signal Characteristics

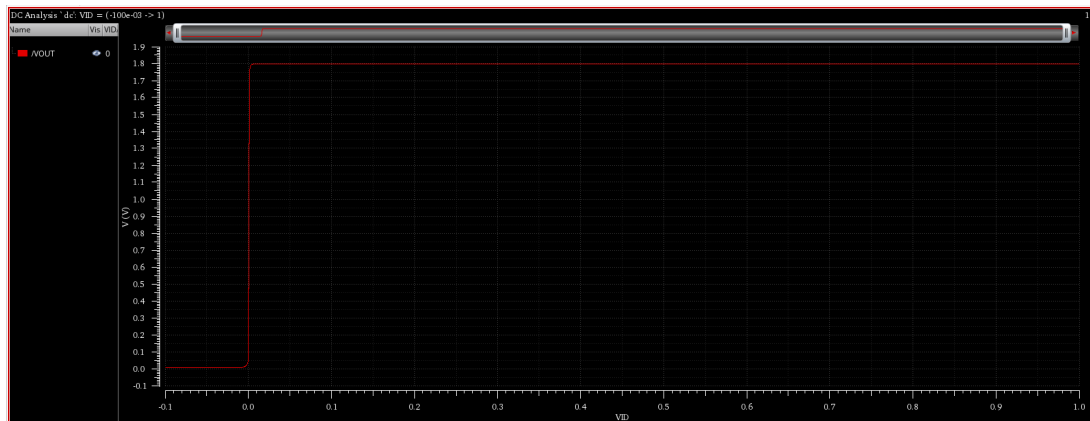


Figure 25: V_{OUT} vs. V_{ID} (DC sweep)

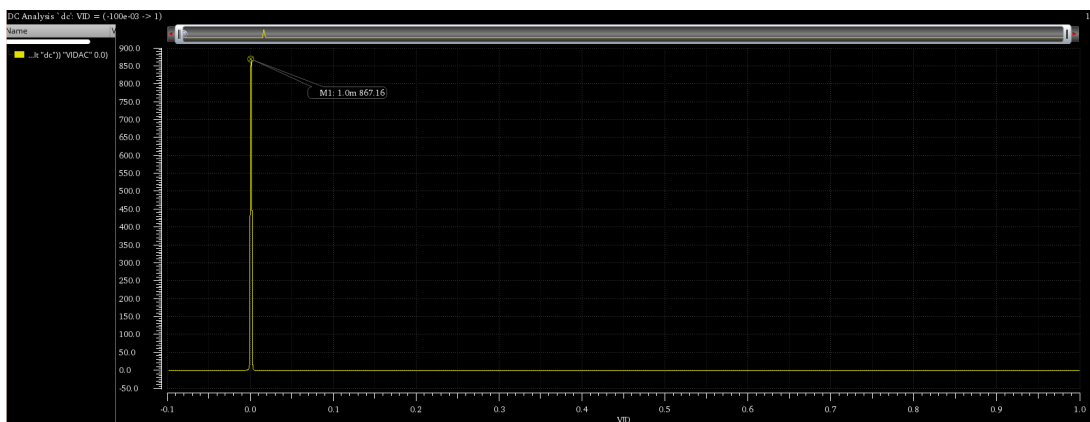


Figure 26: Derivative of V_{OUT} vs. V_{ID}

Large-Signal Diff. Sweep Questions

Q: What is V_{OUT} at $V_{ID} = 0$? Compare with DC OP.

$V_{OUT} = 897.748 \text{ mV}$

Q: Is the peak of the derivative less than A_{vd} from AC analysis? Why?

Yes, because the gain is so large that the 1mV V_{ID} step is not enough. I would use step input differential of 0.1mV

Common-Mode Large-Signal Characteristics (Region vs. VICM)

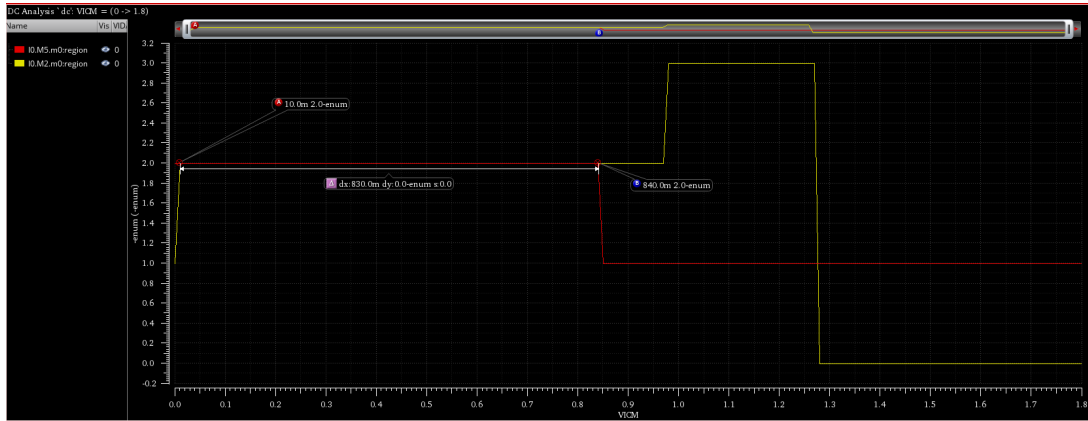


Figure 27: Region parameter vs. VICM for input pair and tail current source

Hand Analysis: Common-Mode Input Range (CMIR)

The Common-Mode Input Range (CMIR) is defined by the range of input voltages that keep all transistors in the first stage operating in the saturation region.

1. Lower Limit ($CMIR_{low}$):

The lowest allowed common-mode input voltage is dictated by the PMOS input pair (M_2, M_3) falling out of saturation. To keep M_2 saturated, its drain-to-source voltage must be greater than its overdrive voltage:

$$\begin{aligned} V_{sd2} &\geq V_{sg2} - |V_{th2}| \\ V_{s2} - V_{d2} &\geq V_{s2} - V_{g2} - |V_{th2}| \\ V_{g2} &\geq V_{d2} - |V_{th2}| \end{aligned}$$

Because the drain of M_2 is tied to the diode-connected NMOS load (M_0), $V_{d2} = V_{gs0}$. Substituting this gives the absolute minimum common-mode voltage:

$$\begin{aligned} CMIR_{low} &= V_{gs0} - |V_{th2,3}| \\ CMIR_{low} &= 733.5 \text{ mV} - 781.6 \text{ mV} \\ CMIR_{low} &= -48.1 \text{ mV} \approx -48 \text{ mV} \end{aligned}$$

2. Upper Limit ($CMIR_{high}$):

The highest allowed common-mode input voltage is restricted by the PMOS tail current source (M_5) being pushed into the triode region. To keep M_5 saturated:

$$\begin{aligned} V_{sd5} &\geq |v_{dsat5}| \\ V_{DD} - V_{s2} &\geq |v_{dsat5}| \\ V_{s2} &\leq V_{DD} - |v_{dsat5}| \end{aligned}$$

Since the source voltage of the input pair is $V_{s2} = V_{g2} + V_{sg2}$, we can substitute this

to find the maximum common-mode input voltage (V_{g2}):

$$V_{g2} + V_{sg2} \leq V_{DD} - |v_{dsat5}|$$

$$CMIR_{high} = V_{DD} - |v_{dsat5}| - V_{sg2,3}$$

Using the extracted $V_{sg2,3}$ and the required saturation margin for the tail source ($|v_{dsat5}| \approx 141.2 \text{ mV}$):

$$CMIR_{high} = 1.8 \text{ V} - 141.2 \text{ mV} - 825.3 \text{ mV}$$

$$CMIR_{high} = 833.5 \text{ mV}$$

CMIR from Region Sweep

Parameter	Hand Analysis	Simulation
CMIR – low	-48m	10m
CMIR – high	833.5m	840m

(Optional) CM Large-Signal (GBW vs. VICM)

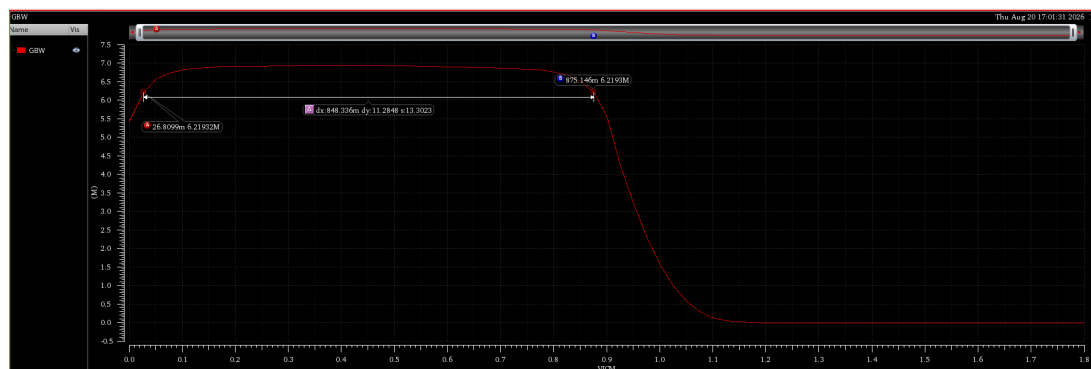


Figure 28: GBW vs. VICM (parametric sweep)

CMIR from GBW Criterion (10% Reduction)

$$26.8 \text{ mV} \rightarrow 875.146 \text{ mV}$$

4. Part 4: Closed-Loop OTA Simulation

4.1 Schematic & DC Operating Point (Unity-Gain Buffer)

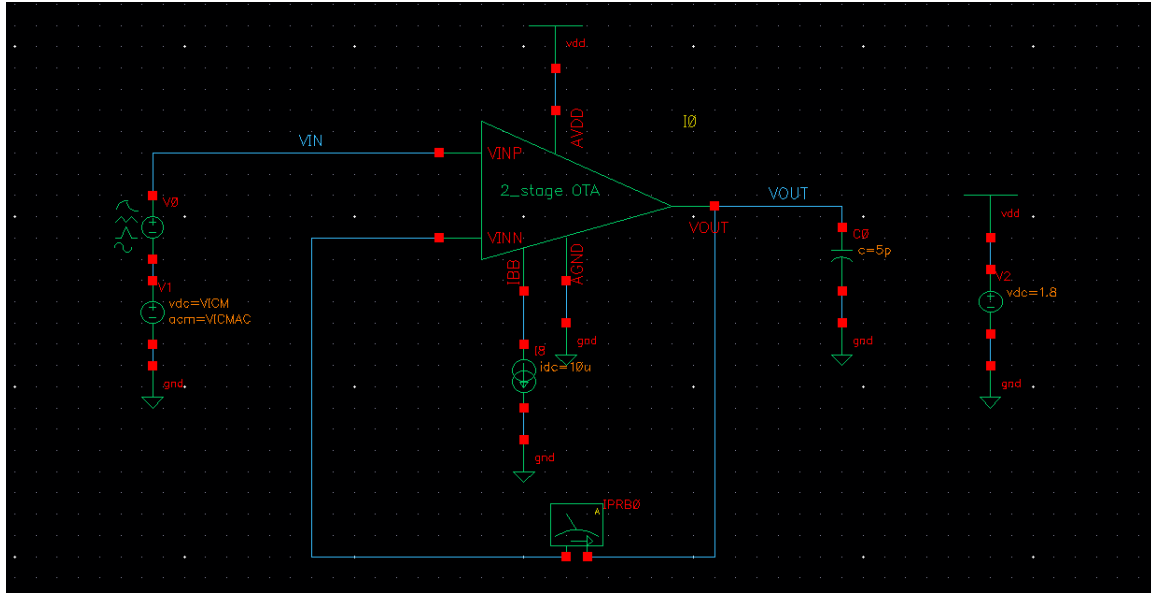


Figure 29: Two-stage Miller OTA in unity-gain buffer configuration

```

1 *****
2 Operating-Point information `dump_op`:
3 *****
4 Operating point computed in info analysis `dump_op' at T = 27 C.
5
6 Transistor Type Region id vgs vds vdsat vth gm gds
7 -----
8 IO.M0.m0 NMOS sat 4.88317 uA 733.5 mV 733.5 mV 99.1935 mV 709.536 mV 78.0618 uS 558.452 nS
9 IO.M1.m0 NMOS sat 4.88598 uA 733.5 mV 738.536 mV 99.1972 mV 709.529 mV 78.0968 uS 557.425 nS
10 IO.M2.m0 PMOS sat -4.88442 uA -825.267 mV -691.822 mV -98.4936 mV -781.58 mV 78.6614 uS 553.32 nS
11 IO.M3.m0 PMOS sat -4.88598 uA -825.322 mV -686.786 mV -98.5179 mV -781.592 mV 78.6749 uS 555.181 nS
12 IO.M4.m0 PMOS sat -10 uA -914.072 mV -914.072 mV -139.197 mV -787.329 mV 114.292 uS 314.589 nS
13 IO.M5.m0 PMOS sat -9.76915 uA -914.072 mV -374.678 mV -139.179 mV -787.355 mV 111.601 uS 665.648 nS
14 IO.M6.m0 PMOS sat -40.3272 uA -914.072 mV -1.19994 V -139.206 mV -787.315 mV 460.669 uS 1.04829 uS
15 IO.M7.m0 NMOS sat 40.3272 uA 738.536 mV 600.055 mV 114.524 mV 685.432 mV 571.878 uS 8.99047 uS
16 IO.M11.m0 NMOS sat 4.85138 uA 733.817 mV 733.817 mV 112.244 mV 684.571 mV 69.7248 uS 1.03118 uS
17 IO.M12.m0 NMOS sat 4.85138 uA 733.817 mV 733.817 mV 112.244 mV 684.571 mV 69.7248 uS 1.03118 uS
18 IO.M13.m0 PMOS sat -4.85138 uA -914.072 mV -332.367 mV -138.312 mV -788.339 mV 55.8276 uS 390.179 nS
19 IO.M14.m0 NMOS triode 22.1699 fA 729.097 mV 44.7059 pV 106.751 mV 689.291 mV 233.256 fS 495.906 uS
20

```

Figure 30: Closed-loop DC operating points at VSIG=VDD/3

Closed-Loop DC OP Questions

Q: Are the DC voltages at the input terminals of the op-amp exactly equal? Why?

No. While negative feedback works to minimize the error signal, a finite open-loop gain means a small residual steady-state error will always exist.

Given the applied input signal $V_{SIG} = 600 \text{ mV}$ and the simulated closed-loop voltage at the inverting terminal $V_{INN} = 600.055 \text{ mV}$, the resulting error voltage is:

$$V_{err} = V_{INN} - V_{SIG}$$

$$V_{err} = 600.055 \text{ mV} - 600 \text{ mV} = 55 \mu\text{V}$$

Q: Is the DC voltage at the output of the first stage exactly equal to the open-loop value? Why?

No, they aren't equal. When the negative feedback loop forces the final output voltage ($V_{\text{out}2}$) to 600 mV, it alters the drain-to-source voltages across the second stage. This shift causes the second-stage branch current (I_{B2}) to increase. To accommodate and sink this higher current, the required gate-to-source voltage of the second-stage driver must increase. Consequently, the output voltage of the first stage ($V_{\text{out}1}$) is forced to rise to maintain this new steady-state condition.

$$V_{\text{out}1,\text{OL}} = 733.735 \text{ mV}$$

$$V_{\text{out}1,\text{CL}} = 738.536 \text{ mV}$$

$$\Delta V_{\text{out}1} = 4.801 \text{ mV}$$

Q: Is the current (and g_m) in the input pair exactly equal? Why?

No, they are not equal because the DC voltage at the input terminals aren't exactly the same.

4.2 Loop Gain (STB Analysis)

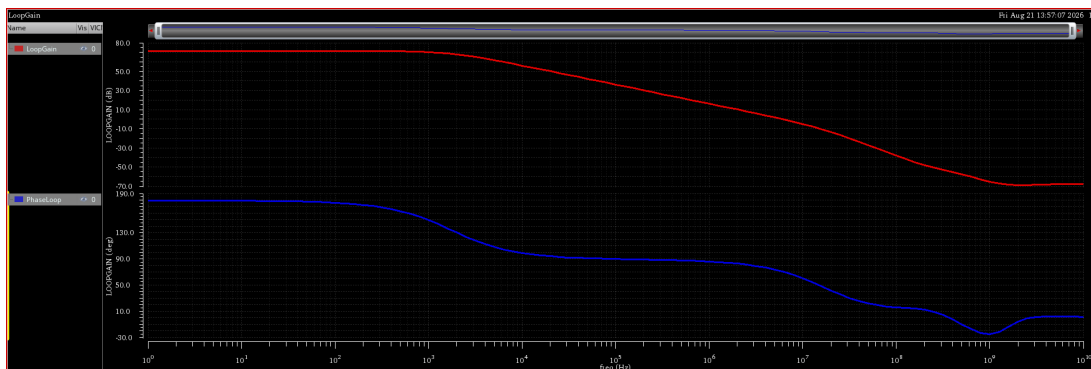


Figure 31: Loop gain magnitude (dB) and phase vs. frequency

Hand Analysis: Second Pole & Phase Margin

1. Non-Dominant Pole (ω_{p2}):

Assuming the right-half-plane zero is perfectly cancelled by the nulling resistor, the second pole is defined by the transconductance of the second stage driver (M_7) and

the load capacitance ($C_L = 5 \text{ pF}$):

$$\begin{aligned}\omega_{p2} &\approx \frac{g_{m7}}{C_L} \\ \omega_{p2} &\approx \frac{571.878 \mu\text{S}}{5 \text{ pF}} \\ \omega_{p2} &\approx 114.375 \times 10^6 \text{ rad/s} \\ f_{p2} &= \frac{\omega_{p2}}{2\pi} \approx 18.2034 \text{ MHz}\end{aligned}$$

2. Phase Margin (PM):

Using a standard two-pole system approximation (where the dominant pole has already shifted the phase by -90°), the Phase Margin depends on how far the gain crossover frequency ($GX \approx GBW = 6.936 \text{ MHz}$) is from the second pole:

$$\begin{aligned}PM &\approx 90^\circ - \tan^{-1} \left(\frac{GX}{f_{p2}} \right) \\ PM &\approx 90^\circ - \tan^{-1} \left(\frac{6.936 \text{ MHz}}{18.2034 \text{ MHz}} \right) \\ PM &\approx 90^\circ - \tan^{-1}(0.381) \\ PM &\approx 90^\circ - 20.858^\circ \\ PM &\approx 69.142^\circ\end{aligned}$$

Loop Gain: Hand Calc vs. Simulation

Parameter	Hand Analysis	Simulation
DC gain (dB)	72.7	72.07
f_u (MHz)	-	6.519
GBW (MHz)	6.98	6.936
Phase Margin ($^\circ$)	69.142	70.03

Q: Compare DC gain, f_u , GBW with the open-loop simulation. Comment.
I dropped parasitic capacitances in w_{p2} and GX calculation.

4.3 Slew Rate

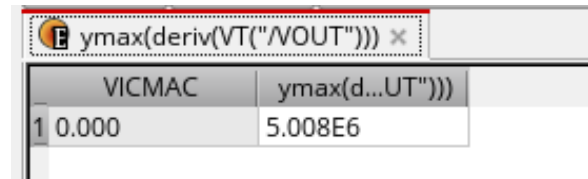
Figure 32: V_{in} and V_{out} overlaid

Figure 33: Slew Rate

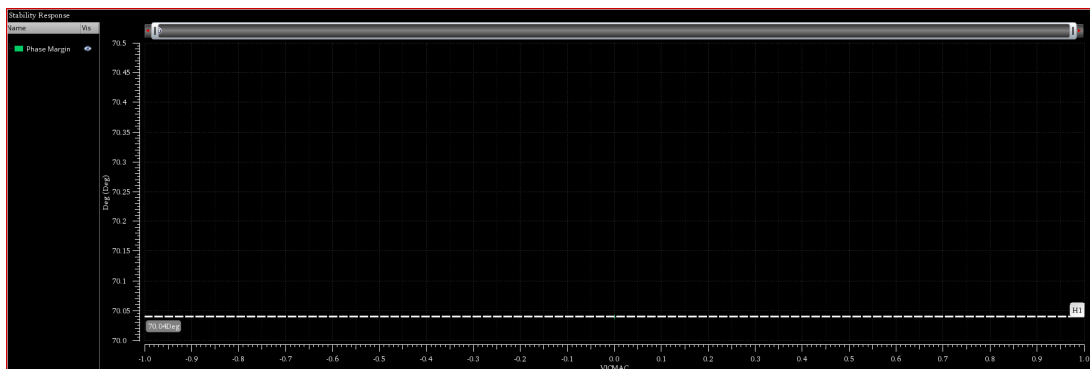


Figure 34: Phase Margin from stb summary

Slew Rate: Hand Calc vs. Simulation

Parameter	Hand Analysis	Simulation
Slew Rate ($V/\mu s$)	5.582	5.008

4.4 Settling Time



Figure 35: Small-signal step response (5 mV step), output rise time

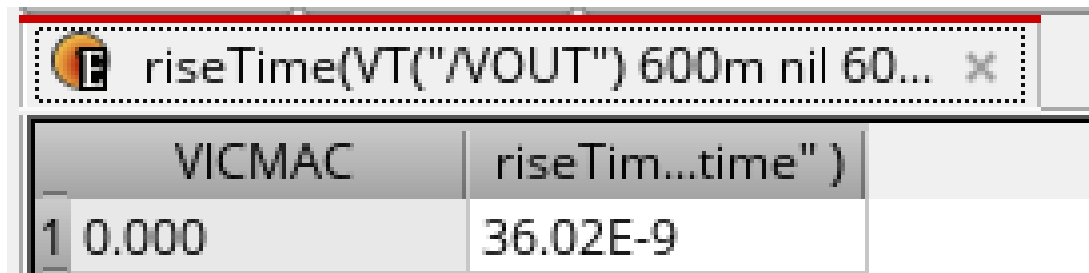


Figure 36: RiseTime

Settling Time: Hand Calc vs. Simulation

Parameter	Hand Analysis	Simulation
Rise time (ns)	53.645	36.02

Q: Do you see any ringing? Why?

No there isn't. It is just a slight overshoot, but Vout settles fast and doesn't swing sinusoidally around the target value.

Q: The simulated result is better than expected—why? (Hint: second-order vs. first-order model)

Second-order systems are generally less damped compared to first-order.

Part 5 (Optional): Verifying RZ tracks 1/gm across corners

Test		Output	Nominal	Spec	Weight	Pass/Fail	Min	Max	C0_0	C0_1	C0_2
MillerOTA:Mill_CL1		1/gm7	1.749k				1.677k	1.826k	1.677k	1.826k	1.749k
MillerOTA:Mill_CL1		ron14	2.006k				1.853k	2.447k	1.853k	2.447k	2.006k

Figure 37: Successful tracking

Test		Output	Nominal	Spec	Weight	Pass/Fail	Min	Max	C0_0	C0_1	C0_2
MillerOTA:Mill_CL1		PM	70.03				69.84	71.36	69.84	71.36	70.03

Figure 38: Phase Margin across corners

Test		Output	Nominal	Spec	Weight	Pass/Fail	Min	Max	C0_0	C0_1	C0_2
MillerOTA:Mill_CL1		SR	5.008M				4.812M	5.077M	5.077M	4.812M	5.008M

Figure 39: Slew Rate across corners

r1m1t_sap.scs		tt		ff		ss		tt		
mosfets.scs										
save1.scs										
Test	Output	Nominal	Spec	Weight	Pass/Fail	Min	Max	C0,0	C0,1	C0,2
MillerOTA:Mill_CL:1	trise	36.02n				32.96n	40.57n	32.96n	40.57n	36.02n

Figure 40: Rise Time across corners

Parameter	Target Value	Achieved
Static gain error	$\leq 0.05\%$	$\sim 0.0232\%$
CMRR @ DC	$\geq 74 \text{ dB}$	84.527 dB
Phase margin	$\geq 70^\circ$	70.04°
Total current consumption(excluding reference and bias current)	$\leq 60 \mu\text{A}$	49.75 μA
CMIR – high	$\geq 0.8 \text{ V}$	0.84 V
CMIR – low	$\leq 0.2 \text{ V}$	0.01 V
Output swing	0.2–1.6 V	0.1127–1.661 V
Buffer rise time (10%–90%)	$\leq 70 \text{ ns}$	36.02 ns
Slew rate	5 V/ μs	5.008 V/ μs
Area	—	106.8327 μm^2
Figure of Merit (FoM)	—	132.9

Table 3: Target vs. Achieved OTA specifications